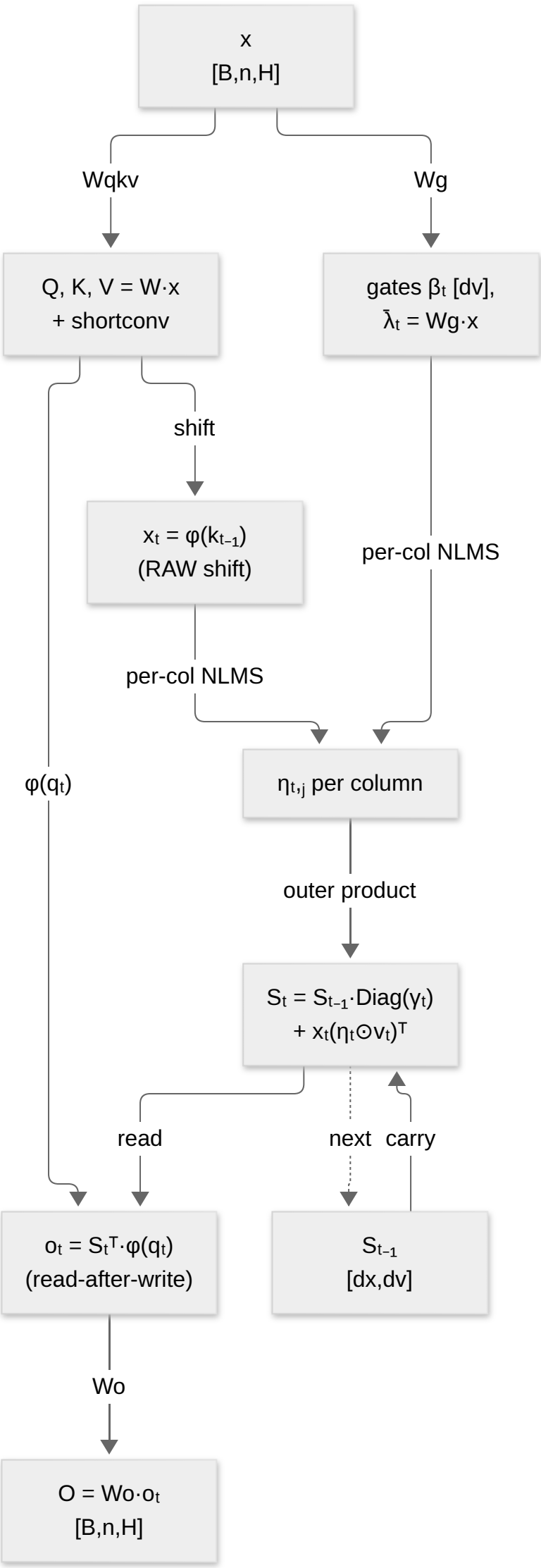

x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates βt [dv],
λ̄t = Wg·x
shift
xt = φ(kt−1)
(RAW shift)
per-col NLMS
per-col NLMS
φ(qt)
ηt,j per column
outer product
St = St−1·Diag(γt)
+ xt(ηt⊙vt)T
read
next
carry
ot = StT·φ(qt)
(read-after-write)
St−1
[dx,dv]
Wo
O = Wo·ot
[B,n,H]